

Question	Answer	Marks
5(a)(i)	amplitude = 0.60 m	A1
5(a)(ii)	oscillations are simple harmonic	B1
5(b)	<i>Any three points from:</i> • mean / equilibrium position is at $h = 1.4$ m • total energy of oscillations = 9.0 J • angular frequency of oscillations = 1.2 rad s^{-1} or period of oscillations = 5.1 s or frequency of oscillation = 0.19 Hz • maximum speed of block = 0.73 m s^{-1} • mass of block = 33 kg	B3
5(c)	U-shaped curve resting on h axis (with minimum at $E_P = 0$)	B1
	curve from $h = 0.8$ m to $h = 2.0$ m, with minimum E_P shown at $h = 1.4$ m	B1
	both end-points of curve shown at $E_P = 9.0$ J	B1

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